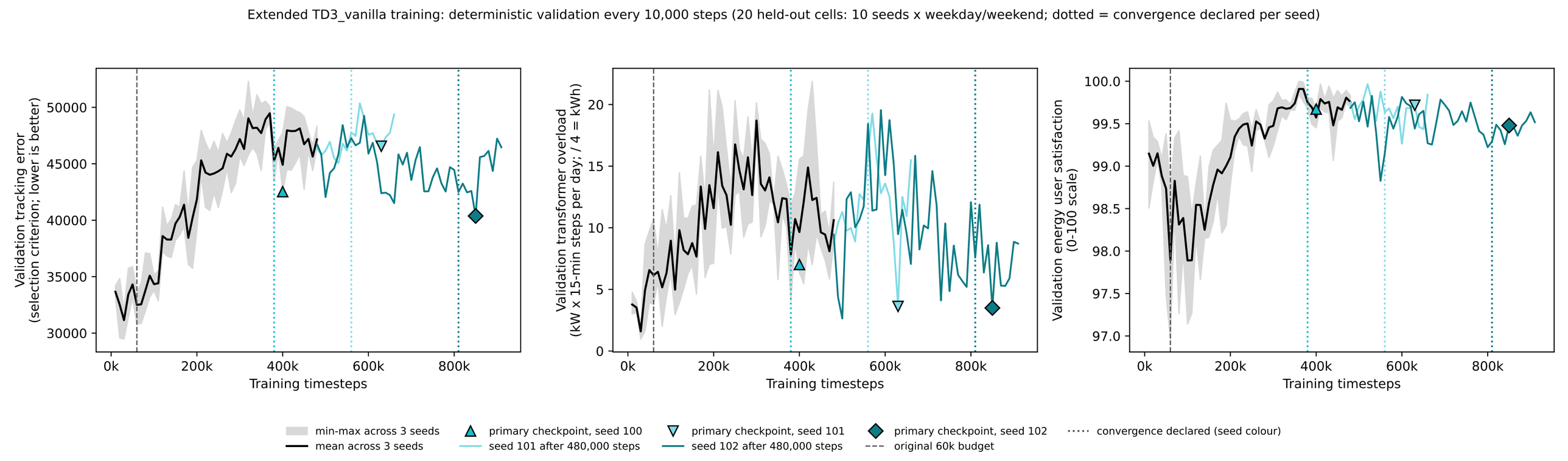
Extended TD3_vanilla training: deterministic validation every 10,000 steps (20 held-out cells: 10 seeds x weekday/weekend; dotted = convergence declared per seed)
Validation tracking error
(selection criterion; lower is better)
50000
45000
40000
35000
30000
0k
200k
400k
600k
800k
Training timesteps
Validation transformer overload
(kW x 15-min steps per day; / 4 = kWh)
20
15
10
5
0
0k
200k
400k
600k
800k
Training timesteps
Validation energy user satisfaction
(0-100 scale)
100.0
99.5
99.0
98.5
98.0
97.5
97.0
0k
200k
400k
600k
800k
Training timesteps
min-max across 3 seeds
mean across 3 seeds
primary checkpoint, seed 100
seed 101 after 480,000 steps
primary checkpoint, seed 101
seed 102 after 480,000 steps
primary checkpoint, seed 102
original 60k budget
convergence declared (seed colour)